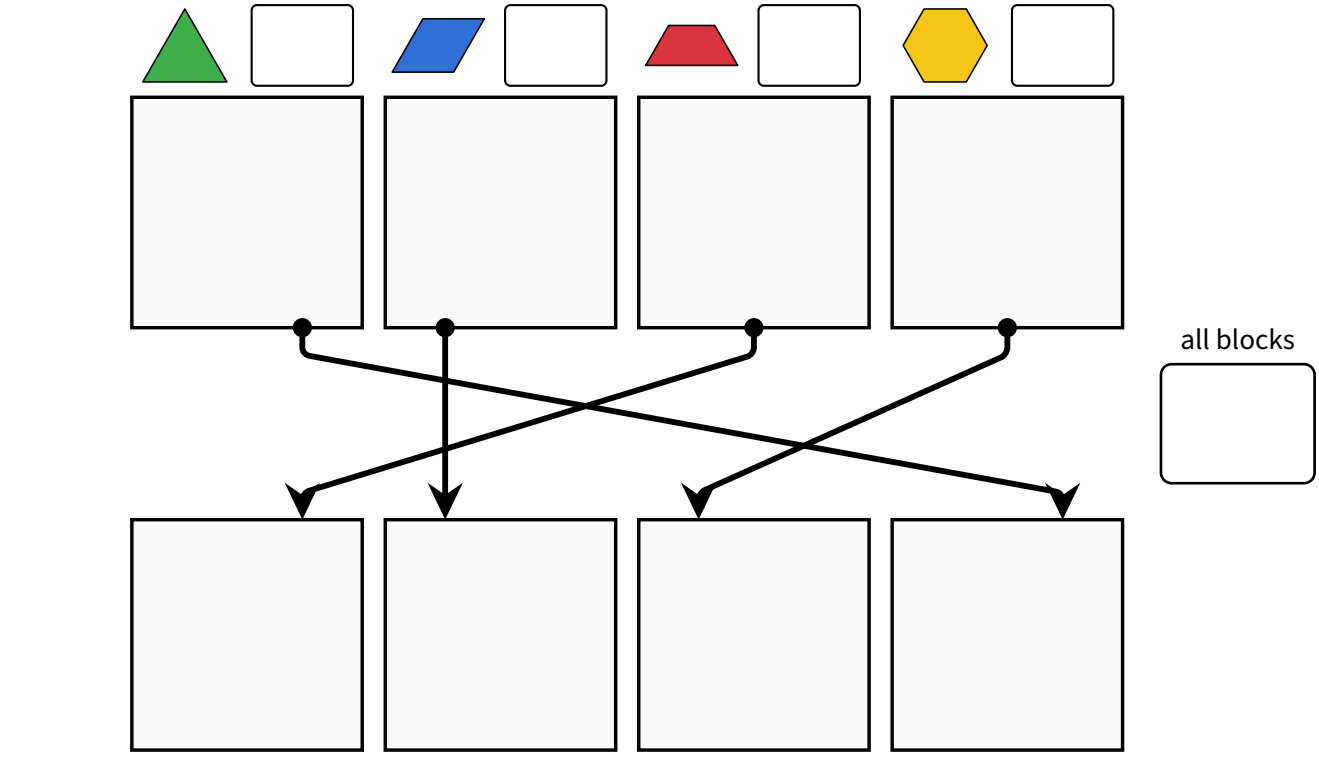
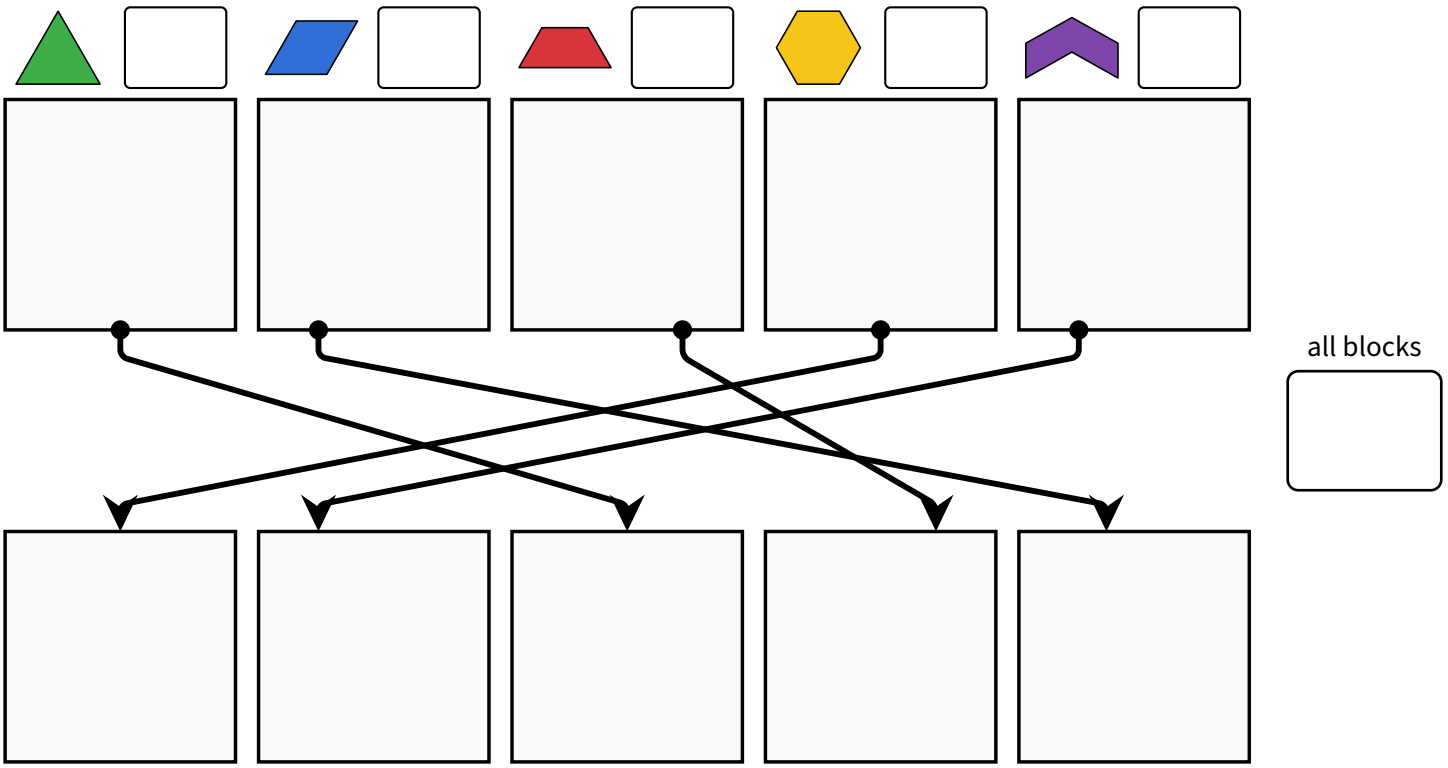
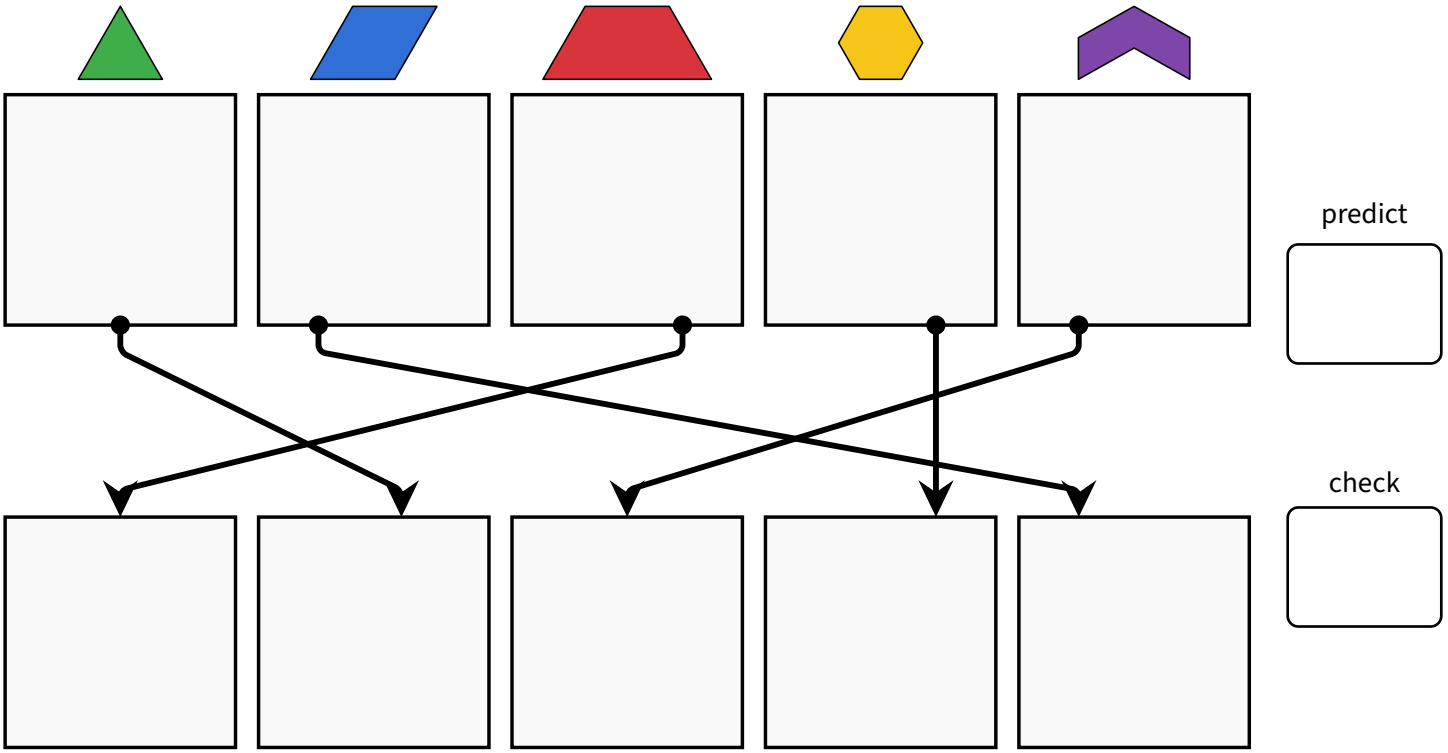
One turn: move every block down its own arrow, then slide the whole bottom row straight up into the top row. When you draw a machine, give every top slot one arrow going out and every bottom slot one arrow coming in. A block is home when it is back under its own picture.

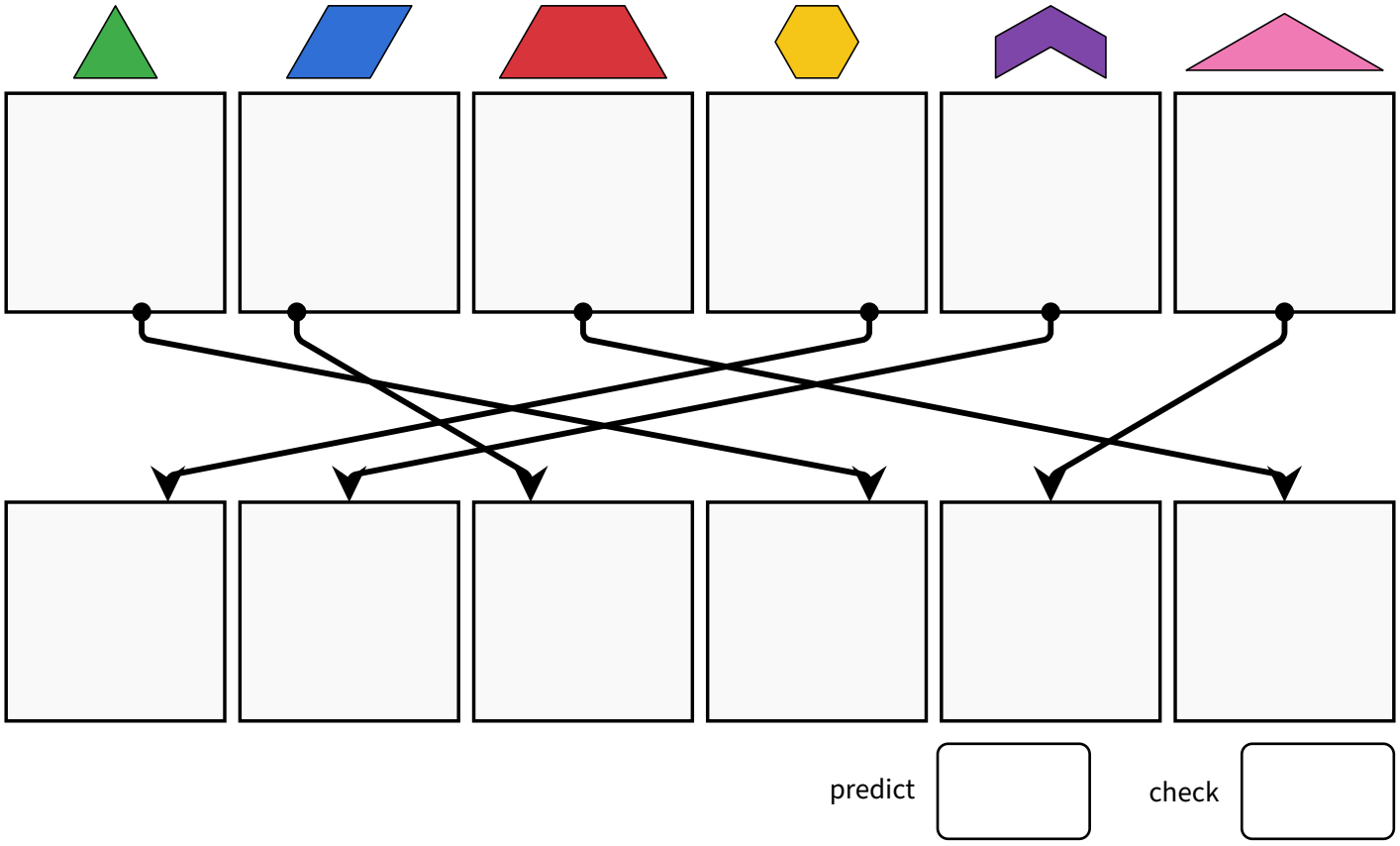
Problem 1: Put each block under its picture and run each machine. For each block, find the first turn after which it is back home. Then find the first turn after which all the blocks are back home at the same time.

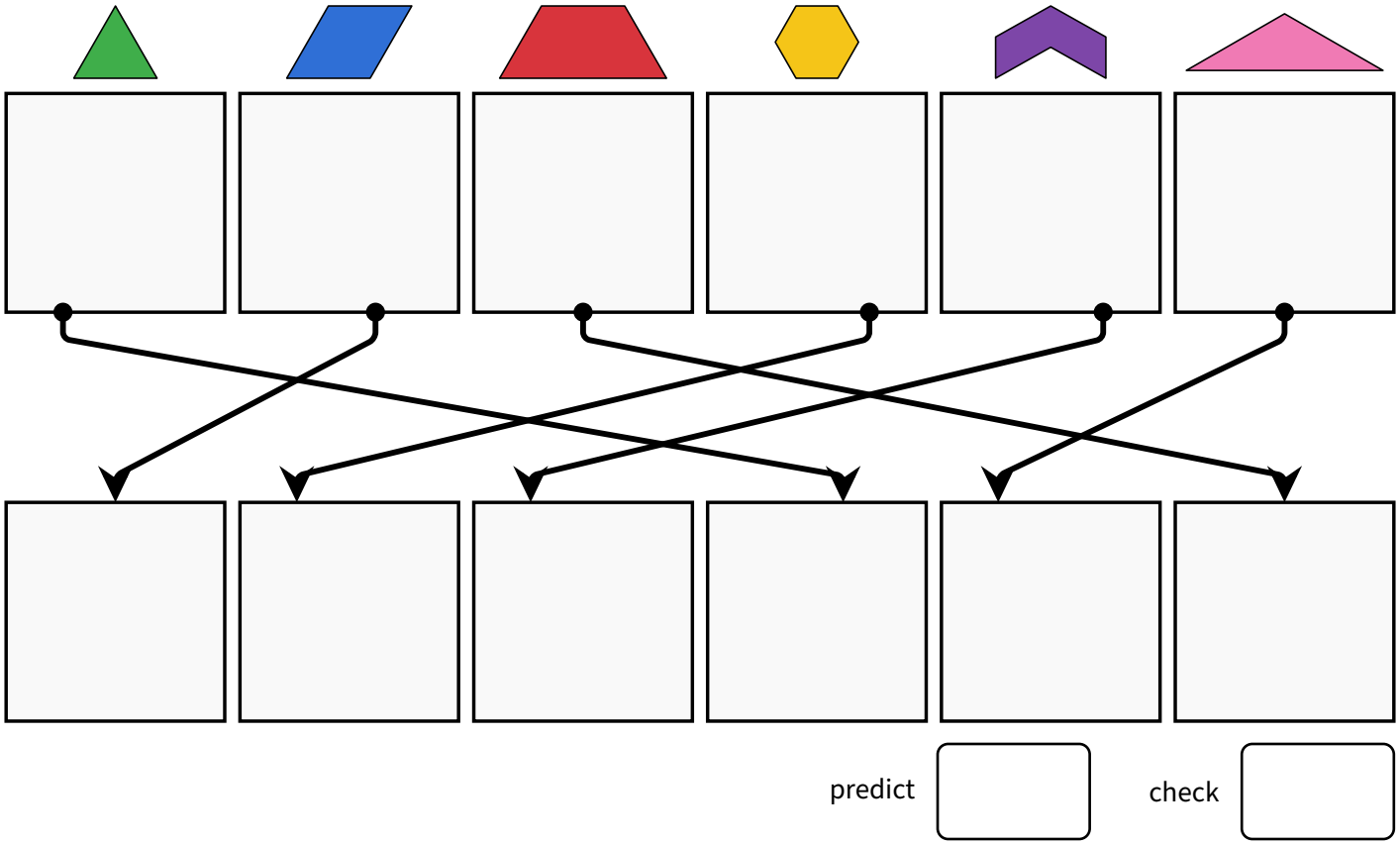


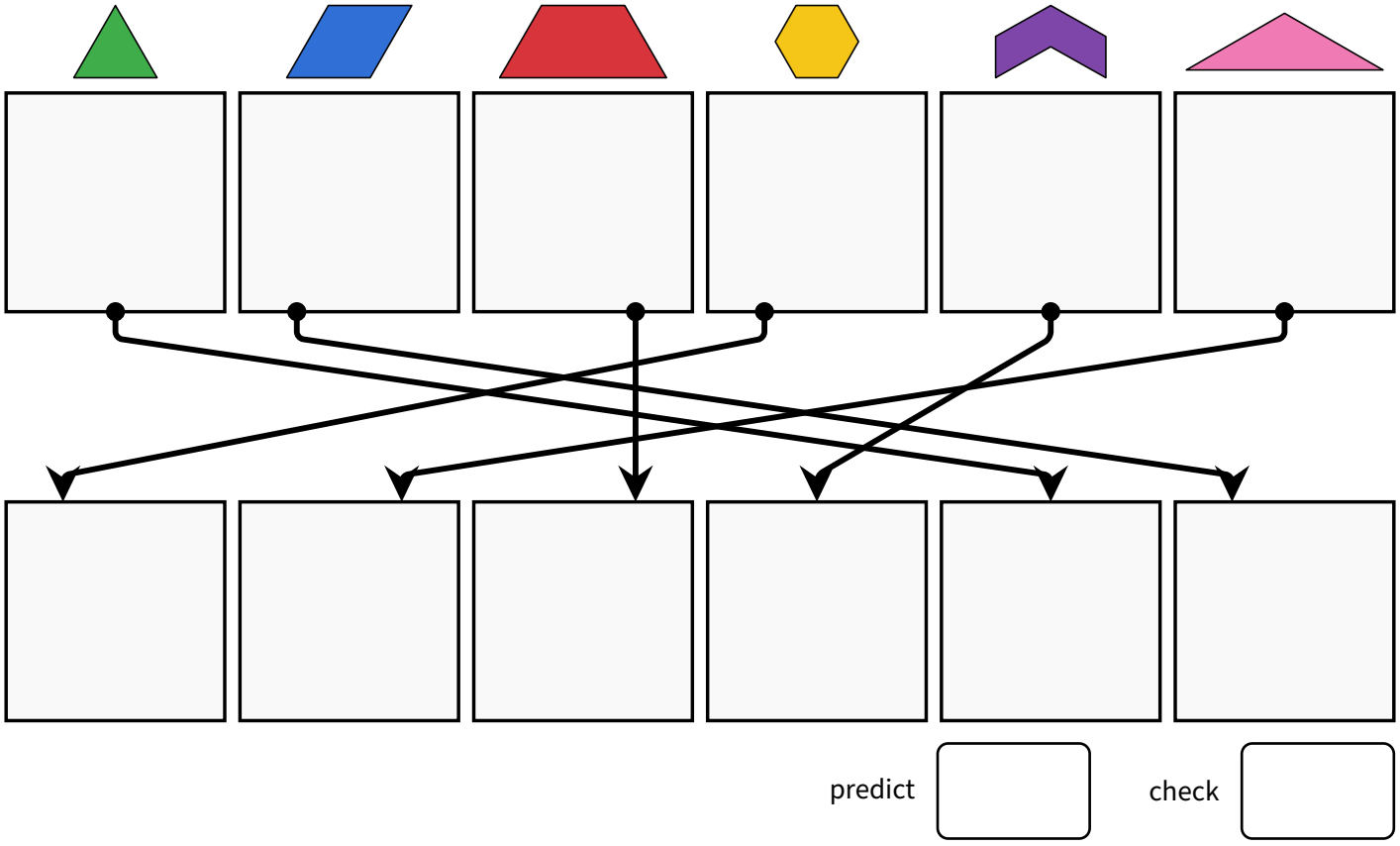


Problem 2: A block's own slot and the slots it visits on its way home make a loop. Without moving any blocks, colour the top slots so that each loop has its own colour, and predict how many turns the machine takes to bring all the blocks home. Then run it to check. Do this for each of the four machines.



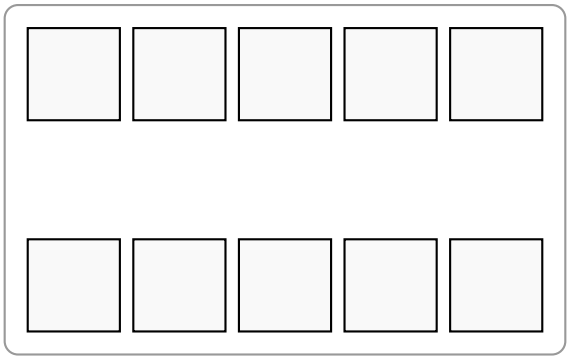



predict
check

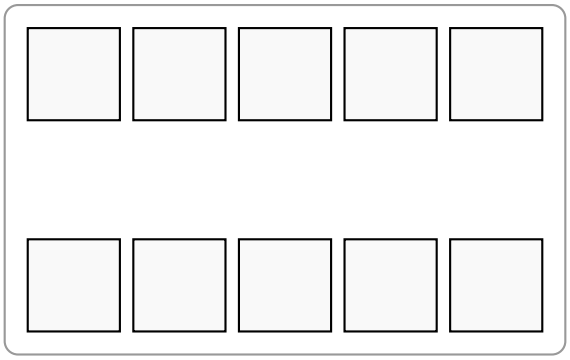

predict
check

Problem 3: For each number of turns from 1 to 6, draw a machine with 5 slots that takes exactly that many turns to bring all the blocks home.

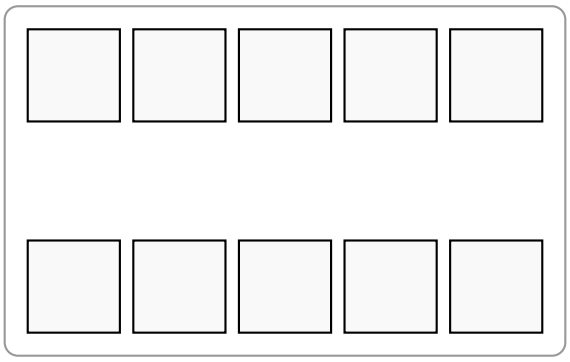
1 turn



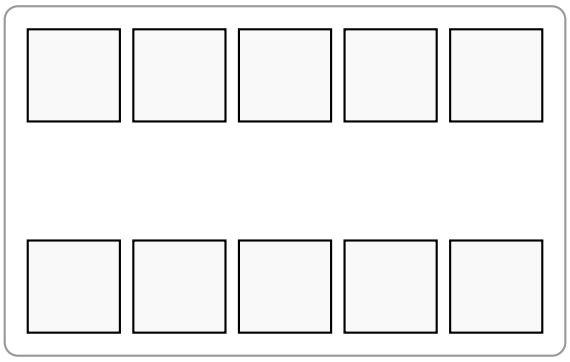
2 turns



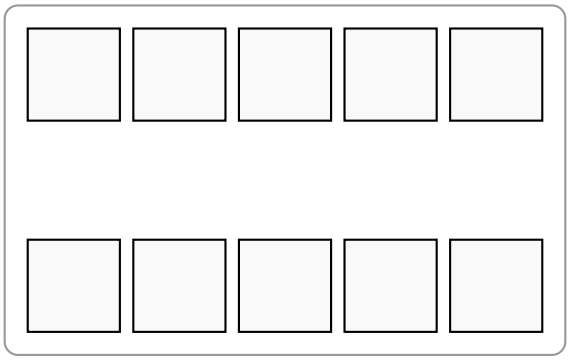
3 turns



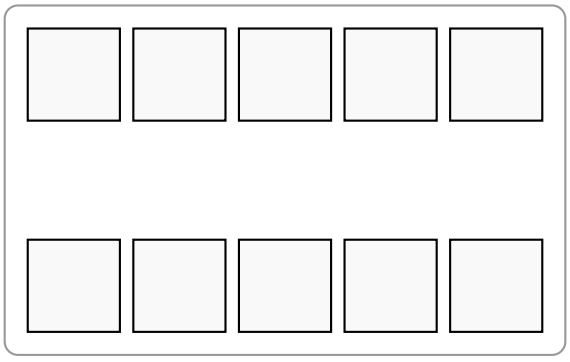
4 turns



5 turns



6 turns



Problem 4: Put each block under its picture on a machine with arrows and do one turn. Move the row of blocks to the empty machine below it without changing its order. Draw arrows on the empty machine so that one turn of it puts every block home. Do this for both machines with arrows.

